

Time-Block Forecasting and Scoring: An Implementation Specification

Research Technical Note: `time_block_forecast`

Spatial Skew- t Simulation Study

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1 Scope and notation

- This note specifies, equation by equation, the computations in `run-settings.R` (block fit and forecast) and `scores.R` (lead-time scoring) of the `time_block_forecast` study. Notation follows the companion note `ar2_rethink`.
- Sites are $s \in \mathcal{S}$ with $n_s = |\mathcal{S}|$, all observed; calendar time is $t \in \{1, \dots, n_t\}$; latent knots are $k \in \{1, \dots, K\}$.
- With $K = 1$ the Voronoi membership $g(s, t) \equiv 1$ is constant, so a single knot serves every site and the knot index is suppressed where harmless.
- Posterior draws are indexed $m \in \{1, \dots, M\}$, blocks $b \in \{1, \dots, B\}$, replicate datasets $d \in \{1, \dots, D\}$.
- Write X_t generically for a latent *Gaussian* process (τ^* or z^*); the observable (τ, z) follow from X by the copula transforms of (??)–(??).

2 Block geometry and the train–test split

- The record carries B seam times $T_o^{(1)} < \dots < T_o^{(B)}$ and one fixed horizon H ; block b partitions time into a contiguous train window and a forecast window

$$\mathcal{T}_b^{\text{tr}} = \{1, \dots, T_o^{(b)}\}, \quad \mathcal{T}_b^{\text{te}} = \{T_o^{(b)} + h : h = 1, \dots, H\}. \quad (1)$$

- The lead time h and the calendar time are linked by $t = T_o^{(b)} + h$; the study fixes

$$n_t = 200, \quad B = 5, \quad (T_o^{(b)})_{b=1}^5 = (50, 80, 110, 140, 170), \quad H = 15. \quad (2)$$

- Every site is observed, so the holdout is purely temporal: the training data of block b is $\{y(s, t) : s \in \mathcal{S}, t \in \mathcal{T}_b^{\text{tr}}\}$ and the target is $\{y(s, t) : s \in \mathcal{S}, t \in \mathcal{T}_b^{\text{te}}\}$.
- Adjacent seams are spaced by $T_o^{(b+1)} - T_o^{(b)} = 30 > H$, so no forecast window reaches the next seam and the B per-block scores are treated as near-independent in (??).

3 Posterior fit on the prefix

- For each (b, d, method) the backend draws M samples from the prefix posterior

$$\pi_b(\theta, \{X_{k,t}^*\}_{t \leq T_o^{(b)}}) = \pi(\theta, \{X_{k,t}^*\}_{t \leq T_o^{(b)}} \mid \{y(s, t) : s \in \mathcal{S}, t \leq T_o^{(b)}\}). \quad (3)$$

- Draw m supplies the parameter bundle

$$\theta^{(m)} = (\beta^{(m)}, \lambda^{(m)}, \alpha^{(m)}, \beta_\tau^{(m)}, \rho^{(m)}, \nu^{(m)}, \gamma^{(m)}, \phi_\tau^{(m)}, \phi_z^{(m)}) \quad (4)$$

together with the imputed observable latent trajectories $\tau_{k,t}^{(m)}, z_{k,t}^{(m)}$ for $t \leq T_o^{(b)}$.

- The gamma copula uses the halved hyperparameters $a^{(m)} = \alpha^{(m)}/2$ and $b^{(m)} = \beta_\tau^{(m)}/2$, matching the generator’s reparameterization.
- The i.i.d. method fixes $\phi_\tau = \phi_z = \mathbf{0}$ and disables temporal pooling; the AR(2) method estimates $\phi_\tau^{(m)}, \phi_z^{(m)} \in \mathbb{R}^2$.

4 Forecasting the held-out window (`forecast_block`)

4.1 Seam-state extraction

- The recursion runs on the Gaussian scale, so the imputed *observable* seam values at $t \in \{T_o - 1, T_o\}$ are mapped back by the forward copula transforms

$$\tau_{k,t}^{*(m)} = \Phi^{-1}(G_{a^{(m)}, b^{(m)}}(\tau_{k,t}^{(m)})), \quad (5)$$

$$z_{k,t}^{*(m)} = \Phi^{-1}(H_{\sigma_{z,t}^{(m)}}(z_{k,t}^{(m)})), \quad \sigma_{z,t}^{(m)} = (\tau_{k,t}^{(m)})^{-1/2}, \quad (6)$$

with Φ the standard normal CDF, $G_{a,b}$ the gamma CDF, and H_σ the half-normal CDF of scale σ .

- The seam state $(X_{k, T_o-1}^{*(m)}, X_{k, T_o}^{*(m)})$ is taken from draw m ’s own imputed trajectory, never from the stationary law $\mathcal{N}(\mathbf{0}, \Sigma)$; by Corollary 1 of the companion note that choice is what preserves the AR(2) signal.

4.2 Latent AR(2) recursion

- For draw m , knot k , and process $X \in \{\tau^*, z^*\}$ with its own coefficients $\phi^{(m)} = (\phi_1^{(m)}, \phi_2^{(m)})$, the window is recursed for $h = 1, \dots, H$:

$$X_{k, T_o+h}^{*(m)} = \phi_1^{(m)} X_{k, T_o+h-1}^{*(m)} + \phi_2^{(m)} X_{k, T_o+h-2}^{*(m)} + \sigma^{(m)} \xi_{k,h}^{(m)}, \quad \xi_{k,h}^{(m)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (7)$$

- The innovation standard deviation is fixed by Yule–Walker at unit marginal variance ($\gamma_0 = 1$), so the latent variance is preserved across the forecast:

$$\gamma_1 = \frac{\phi_1^{(m)}}{1 - \phi_2^{(m)}}, \quad \gamma_2 = \phi_1^{(m)} \gamma_1 + \phi_2^{(m)}, \quad (\sigma^{(m)})^2 = 1 - \phi_1^{(m)} \gamma_1 - \phi_2^{(m)} \gamma_2. \quad (8)$$

- A draw whose $\phi^{(m)}$ leaves the stationarity triangle $\{|\phi_2| < 1, \phi_1 + \phi_2 < 1, \phi_2 - \phi_1 < 1\}$ is clamped to $\sigma^{(m)} = 1$, capping the non-stationary recursion at the i.i.d. marginal.

4.3 The i.i.d. baseline

- The i.i.d. method discards the seam state and draws each latent forecast slice fresh from the stationary marginal:

$$\tau_{k,T_o+h}^{\star(m)}, z_{k,T_o+h}^{\star(m)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad h = 1, \dots, H. \quad (9)$$

- Equation (??) is exactly the $h \rightarrow \infty$ limit of (??), so the two methods differ *only* through the seam-state conditioning.

4.4 Copula transforms and the spatial draw

- After the recursion the latent window is mapped to the observable scale by the inverse copula transforms

$$\tau_{k,T_o+h}^{(m)} = G_{a^{(m)}, b^{(m)}}^{-1}(\Phi(\tau_{k,T_o+h}^{\star(m)})), \quad (10)$$

$$z_{k,T_o+h}^{(m)} = H_{\sigma_z^{(m)}}^{-1}(\Phi(z_{k,T_o+h}^{\star(m)})), \quad \sigma_z^{(m)} = (\tau_{k,T_o+h}^{(m)})^{-1/2}, \quad (11)$$

which invert (??)–(??) and apply *after* the recursion.

- The τ – z coupling is enforced within the draw: $\sigma_z^{(m)}$ in (??) uses the same draw’s forecasted $\tau_{k,T_o+h}^{(m)}$, not an independent marginal.
- For each lead h (time $t = T_o + h$) the spatial field is drawn from the skew- t observation model with $K = 1$ membership $g \equiv 1$:

$$\hat{y}^{(m)}(s, t) = x(s, t)^\top \beta^{(m)} + \lambda^{(m)} z_{1,t}^{(m)} + \varepsilon^{(m)}(s, t), \quad (12)$$

$$\varepsilon^{(m)}(\cdot, t) \sim \mathcal{N}(\mathbf{0}, (\tau_{1,t}^{(m)})^{-1} C^{(m)}), \quad C_{ss'}^{(m)} = C(s, s'; \rho^{(m)}, \nu^{(m)}, \gamma^{(m)}). \quad (13)$$

- Operationally (??) is realized as $\varepsilon^{(m)}(\cdot, t) = L^{(m)} \zeta$, with $L^{(m)} L^{(m)\top} = C^{(m)}$ the Cholesky factor and $\zeta \sim \mathcal{N}(\mathbf{0}, (\tau_{1,t}^{(m)})^{-1} I)$.
- The block output is the predictive sample array $\{\hat{y}^{(m)}(s, T_o^{(b)} + h)\}$ of shape $M \times n_s \times H$, paired with the validation truth $y_b^{\text{val}}(s, h) = y(s, T_o^{(b)} + h)$.
- Averaging over $m = 1, \dots, M$ integrates (??) and the seam state jointly, so the predictive carries both parameter and state uncertainty with no plug-in of posterior means.

5 Score computation (scores.R)

- Each (b, s, h) predictive is the M -point empirical measure

$$\hat{F}_{b,s,h} = \frac{1}{M} \sum_{m=1}^M \delta_{\hat{y}^{(m)}(s, T_o^{(b)} + h)}, \quad (14)$$

and all scores below are functionals of $\hat{F}_{b,s,h}$ and the truth $y_b^{\text{val}}(s, h)$.

5.1 Lead-time CRPS

- The per-site CRPS uses the empirical-distribution estimator

$$\text{CRPS}(\hat{F}_{b,s,h}, y) = \frac{1}{M} \sum_{m=1}^M |\hat{y}^{(m)} - y| - \frac{1}{2M^2} \sum_{m=1}^M \sum_{m'=1}^M |\hat{y}^{(m)} - \hat{y}^{(m')}|, \quad (15)$$

with $\hat{y}^{(m)} = \hat{y}^{(m)}(s, T_o^{(b)} + h)$; this is the $d = 1$ case of the energy score.

- The block- b , lead- h score is the site average

$$S^{\text{CRPS}}(h; b, d) = \frac{1}{n_s} \sum_{s \in \mathcal{S}} \text{CRPS}(\hat{F}_{b,s,h}, y_b^{\text{val}}(s, h)). \quad (16)$$

5.2 Lead-time Brier

- Thresholds are the empirical quantiles of the block's validation field at the probability grid $\mathbf{p} = (0.90, 0.92, 0.94, 0.95, 0.96, 0.98, 0.99)$:

$$c_{b,q} = \hat{Q}_{p_q}(\{y(s, t) : s \in \mathcal{S}, t \in \mathcal{T}_b^{\text{te}}\}), \quad q = 1, \dots, |\mathbf{p}|. \quad (17)$$

- The forecast exceedance probability and the lead- h , threshold- q Brier score are

$$\hat{p}_b(s, h; q) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{\hat{y}^{(m)}(s, T_o^{(b)} + h) > c_{b,q}\}, \quad (18)$$

$$S^{\text{BS}}(h, q; b, d) = \frac{1}{n_s} \sum_{s \in \mathcal{S}} \left(\mathbf{1}\{y_b^{\text{val}}(s, h) > c_{b,q}\} - \hat{p}_b(s, h; q) \right)^2. \quad (19)$$

5.3 Joint energy and variogram scores

- For site s the held-out window is the length- H vector $\mathbf{y}_{b,s} = (y(s, T_o^{(b)} + 1), \dots, y(s, T_o^{(b)} + H)) \in \mathbb{R}^H$, with predictive sample $\hat{\mathbf{y}}_{b,s}^{(m)} \in \mathbb{R}^H$.
- The energy score (sample estimator, $\|\cdot\|$ the Euclidean norm on $\mathbb{R}^H$) is

$$\widehat{\text{ES}}(b, s) = \frac{1}{M} \sum_{m=1}^M \|\hat{\mathbf{y}}_{b,s}^{(m)} - \mathbf{y}_{b,s}\| - \frac{1}{2M^2} \sum_{m=1}^M \sum_{m'=1}^M \|\hat{\mathbf{y}}_{b,s}^{(m)} - \hat{\mathbf{y}}_{b,s}^{(m')}\|. \quad (20)$$

- The variogram score of order $p = \frac{1}{2}$, unit weights, and i, j indexing lead time is

$$\widehat{\text{VS}}(b, s) = \sum_{i=1}^H \sum_{j=1}^H \left(|y_{b,s,i} - y_{b,s,j}|^p - \frac{1}{M} \sum_{m=1}^M |\hat{y}_{b,s,i}^{(m)} - \hat{y}_{b,s,j}^{(m)}|^p \right)^2. \quad (21)$$

- Each is reduced to one number per block by site averaging,

$$S^{\text{ES}}(b, d) = \frac{1}{n_s} \sum_s \widehat{\text{ES}}(b, s), \quad S^{\text{VS}}(b, d) = \frac{1}{n_s} \sum_s \widehat{\text{VS}}(b, s). \quad (22)$$

- In (??)–(??) the lead time supplies the multivariate dimension, so ES and VS credit the temporal dependence that the pointwise CRPS of (??) cannot resolve.

5.4 Lead-time curve

- The Brier lead curve first averages (??) over the threshold grid,

$$S^{\text{BS}}(h; b, d) = \frac{1}{|\mathbf{p}|} \sum_{q=1}^{|\mathbf{p}|} S^{\text{BS}}(h, q; b, d). \quad (23)$$

- For a univariate score $S \in \{S^{\text{CRPS}}, S^{\text{BS}}\}$ the per-block mean over the D replicate datasets is

$$\bar{S}_b(h) = \frac{1}{D} \sum_{d=1}^D S(h; b, d). \quad (24)$$

- The lead-time curve and its standard error follow from the dispersion of the B per-block means,

$$\bar{S}(h) = \frac{1}{B} \sum_{b=1}^B \bar{S}_b(h), \quad \text{SE}(h) = \frac{1}{\sqrt{B}} \left(\frac{1}{B-1} \sum_{b=1}^B (\bar{S}_b(h) - \bar{S}(h))^2 \right)^{1/2}. \quad (25)$$

- Because the seam spacing $30 > H$ makes the B summands of (??) near-independent, $\text{SE}(h)$ is a valid uncertainty band for $\bar{S}(h)$.

6 Verification conditions

- **$H = 0$ parity.** Re-scoring time T_o must give the AR(2) and i.i.d. methods the same predictive; any difference is a seam-indexing error in (??)–(??).
- **$\phi = \mathbf{0}$ collapse.** A draw with $\phi^{(m)} = \mathbf{0}$ must reduce (??) to (??) immediately; residual autocorrelation signals a double-counted seam state.
- **Long-horizon limit.** With stationary $\phi^{(m)}$ and $H \gg h^*$, the forecasted latent slices must reproduce the $\mathcal{N}(0, 1)$ marginal, confirming the Yule–Walker normalization (??).